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AI-driven decision support for disaster management and humanitarian logistics: Models, applications, and research challenges

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ABSTRACT

As disasters grow more frequent, complex, and destructive, the demand for intelligent and responsive logistics decision-making has intensified. This systematic review examines the integration of Artificial Intelligence (AI) into Decision Support Systems (DSS) with a focus on disaster response and humanitarian logistics. Drawing from peer-reviewed literature published between 2008 and 2024, the review identifies major AI techniques, including machine learning, deep learning, natural language processing, reinforcement learning, and AI–IoT integration, and evaluates their application in key humanitarian logistics functions such as demand forecasting, facility location and prepositioning, transportation routing, last-mile delivery, and resource allocation under uncertainty. The findings reveal a concentration of research on natural disasters and early-phase emergency response, while recovery, mitigation, and long-term supply chain resilience remain underexplored. Persistent challenges include fragmented and incomplete datasets, limited explainability of AI-driven models, siloed system designs, and ethical concerns around fairness and equity in resource distribution. The review concludes by outlining a research agenda that calls for interoperable, interpretable, and context-sensitive AI-DSS frameworks capable of enhancing supply chain coordination, equity in aid distribution, and decision quality across the full humanitarian logistics lifecycle. This review is the first to systematically categorize existing studies by both AI technique and disaster phase and introduces an integrative conceptual framework that links AI methods to humanitarian logistics decision functions.

1. Introduction

Disasters are increasing in frequency, scale, and complexity, placing extraordinary pressures on global supply chains and humanitarian operations. Climate-induced hazards, pandemics, and man-made crises disrupt the flow of critical goods, create surges in

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demand for relief supplies, and expose structural vulnerabilities in logistics networks (Chase et al., 2025). In 2022 alone, more than 185 million people were affected by major disasters worldwide, generating economic losses exceeding $220 billion (CRED, 2024). Projections suggest that by 2030, the world may face nearly 560 disaster events annually (UNDRR, 2022). These dynamics make disaster logistics one of the most pressing domains within operations and supply chain management, where speed, efficiency, and equity are simultaneously required under severe uncertainty (Altay & Green III, 2006; Comfort et al., 2004; Kovács & Spens, 2007).

Humanitarian logistics differs fundamentally from commercial supply chains in its objectives and constraints. Relief operations must prioritize lives over profits, operate with incomplete information, and function in environments where infrastructure may be damaged or inaccessible. Decision problems span strategic, tactical, and operational levels: prepositioning relief supplies and locating facilities, forecasting demand for affected populations, optimizing transportation routes under uncertainty, and managing last-mile delivery in highly disrupted settings (Holguín-Veras et al., 2012; Kovács & Spens, 2007; Y. Wang & Sarkis, 2021). Traditional decision support tools have improved coordination in such contexts, yet they often lack the agility and adaptability required for large-scale, fast-evolving crises.

These limitations have brought Decision Support Systems (DSS) to the forefront of disaster operations research, especially when paired with recent advances in Artificial Intelligence (AI) (Dargan et al., 2020; Power, 2002). AI-enabled DSS (AI-DSS), extend the capabilities of traditional systems by enabling real-time analytics, pattern recognition, and intelligent automation across vast and heterogeneous data sources, including satellite imagery, social media, and Internet of Things (IoT) networks (Al Marzooqi, 2024; Albahri et al., 2024). Techniques such as machine learning (ML), deep learning (DL), natural language processing (NLP), and computer vision are being applied to predict hazards, support evacuation planning, optimize logistics, and monitor cascading impacts (Akter & Wamba, 2019; Singh et al., 2024). From an Operations Management (OM) perspective, AI-DSS have the potential to transform key disaster management tasks: forecasting risks, optimizing evacuation and logistics, reallocating resources dynamically, and monitoring cascading disruptions (Choi et al., 2022; Mithas et al., 2022). However, much of the existing literature focuses narrowly on technical model performance or specific disaster types, offering limited insight into how these technologies support structured decision processes, especially across the full disaster management lifecycle (Hasanuzzaman et al., 2023). Furthermore, existing reviews often emphasize technical contributions (e.g., algorithm design, model accuracy) or focus narrowly on specific disaster types (e.g., wildfires, pandemics), with limited synthesis of how AI technologies support real-world decisions in complex, uncertain disaster environments (Ibrahim et al., 2025; Kim et al., 2023), factors critical for OM researchers and system designers.

This paper responds to these gaps by offering a structured, decision-oriented review of AI-DSS for disaster management. Rather than cataloging algorithms or use cases in isolation, we propose a conceptual framework that links AI techniques to decision functions and disaster phases. Our goal is to help OM scholars, especially those new to the disaster domain, position their work within a broader operational context and identify high-impact research opportunities. Our contributions are threefold: (1) *Framework Contribution*: We develop an integrative framework derived from systematic classification of over 100 studies, organizing the literature across functional decision types, disaster lifecycle phases, and AI methodologies. (2) *Gap Analysis*: We highlight underexplored areas, such as recovery-phase optimization, explainability in high-stakes contexts, and AI integration across interdependent systems (e.g., health-logistics-energy). (3) *Research Agenda*: We propose actionable future research directions, including promising data sources, modeling approaches, and governance mechanisms for responsible AI-DSS deployment.. To guide our synthesis, we address five research questions:

- RQ1: What types of AI methods have been integrated into decision support systems for disaster management?
- RQ2: In which phases of the disaster management cycle (preparedness, response, recovery, mitigation) are AI-DSS most frequently applied?
- RQ3: What are the main functional domains (e.g., health, logistics, infrastructure) and decision tasks supported by AI-DSS?
- RQ4: What are the challenges in designing, deploying, and governing AI-DSS in real-world disaster contexts?
- RQ5: What are the key research gaps, promising data sources, and methodological innovations needed to advance the field?

By addressing these questions, this review advances humanitarian logistics research by clarifying how AI-driven decision support can improve supply chain agility, efficiency, and equity in disaster settings. Mapping AI methods to logistics functions such as facility location, prepositioning, transportation routing, last-mile delivery, and resource allocation provides a structured basis for integrating emerging technologies with established humanitarian logistics models. Building on prior scholarship in related outlets that underscores the distinctive constraints of humanitarian operations (Altay & Green III, 2006; Holguín-Veras et al., 2012; Kovács & Spens, 2007), this paper positions AI-enabled decision support as a critical enabler of resilient and inclusive humanitarian supply chains, extending the theoretical foundations of humanitarian logistics and offering actionable pathways for practice.

The remainder of this paper is organized as follows. Section 2 defines key concepts and presents the theoretical background. Section 3 details the review methodology. Section 4 introduces the conceptual framework linking AI techniques, decision functions, and disaster phases. Section 5 synthesizes findings across risk forecasting, logistics, recovery, and human-centered AI. Section 6 discusses data and modeling considerations, followed by Section 7, which outlines research gaps and future directions. Section 8 concludes with theoretical contributions, practical implications, and limitations.

2. Background and definitions

This section introduces the core concepts necessary to contextualize the review, including disaster management phases, the role of DSS, and the integration of AI within these systems. These definitions provide a foundation for subsequent sections' conceptual framework and thematic synthesis.

2.1. Key concepts and terminology

Disaster management refers to the organized process of preparing for, responding to, recovering from, and mitigating disasters through strategic planning and operational execution (Coppola, 2020; FEMA, 2022; UNDRR, 2009). These phases, preparedness, response, recovery, and mitigation, form the disaster management cycle (Alexander, 2002, p. 1985; Bullock et al., 2017; Settle, 1985). Each phase presents distinct decision-making challenges and data needs, which AI-DSS increasingly aims to address. As shown in Fig. 1, natural disasters, such as floods, earthquakes, and wildfires, occur more frequently than technological or man-made disasters, and their frequency is increasing due to climate change and urbanization (CRED, 2024; UNDRR, 2022). Man-made disasters, such as pandemics, cyberattacks, and industrial accidents, pose unique challenges due to their unpredictability, data sensitivity, and socio-political complexity (F. I. Khan & Abbasi, 1999; Smith, 2013; WHO, 2023). DSS are interactive, computer-based platforms that assist with unstructured or semi-structured decisions by integrating data, models, and user interfaces (Power, 2002; Shim et al., 2002). In disaster management, DSS have evolved from GIS-based mapping and simulation tools to dynamic platforms integrating real-time inputs (Aerts et al., 2018; Altay & Green III, 2006). AI augments DSS with advanced capabilities in perception, prediction, and decision-making. As summarized in Table 1, key AI techniques used in disaster contexts include:

- AI refers to the field of computer science focused on building systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, perception, and language understanding (Russell & Norvig, 2021).
- ML: Algorithms that learn from data to make predictions (e.g., flood forecasting, damage estimation) (Bhatt & Zaveri, 2002).
- Deep Learning (DL): A subfield of ML using neural networks for high-dimensional data such as images or video (e.g., wildfire detection from satellite imagery) (Jain et al., 2020).
- NLP: AI methods for analyzing human language, useful in extracting situational awareness from social media or emergency calls (Imran et al., 2015).
- Computer Vision: Enables analysis of images or video feeds, such as assessing building damage or monitoring crowd behavior (Sun et al., 2020).
- Reinforcement Learning (RL): AI systems that learn optimal actions by interacting with their environment, applicable in logistics and resource allocation during disasters (Holguín-Veras et al., 2012; Prashanth & Bhatnagar, 2010).
- Internet of Things (IoT) and AI Integration: Real-time sensing and anomaly detection (A. Khan et al., 2020).

2.2. Historical trajectory and integration of DSS and AI

DSS in disaster management originated with statistical and simulation tools for logistics and hazard planning. The integration of GIS in the 1990 s enabled spatial visualization and loss estimation (Cutter et al., 2003; Zerger & Smith, 2003). However, traditional DSS were limited by rigid rule-based logic and struggled to process unstructured or real-time data. The rise of AI marked a shift toward adaptive, predictive systems capable of ingesting heterogeneous data, satellite feeds, social media, IoT sensor logs, and providing actionable insights in dynamic environments (Albahri et al., 2024; Dargan et al., 2020). Deep learning and NLP expanded DSS functionality to include visual classification and text mining, while hybrid AI-DSS systems increasingly support end-to-end decision pipelines (Imran et al., 2015; Xu et al., 2019). This evolution is summarized in Table 2.

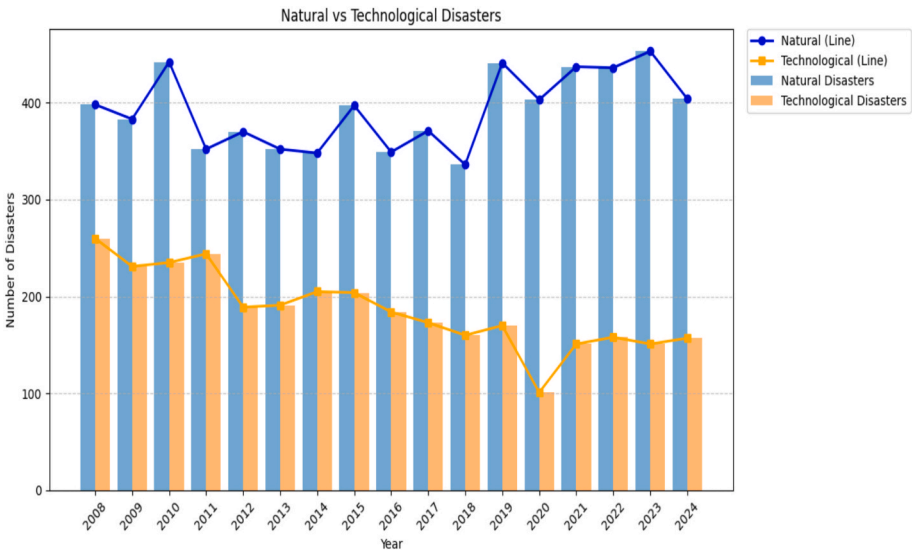

Fig. 1. Natural and Technological disasters Reported Globally (2008–2024) (EM-DAT, 2024).

Table 1
Common AI Techniques and Their Roles in DSS.

AI Technique	Function in Disaster Management	Example	Advantages	Limitations
ML (Mosavi et al., 2018)	Predictive analytics for flood forecasting	SVM, Random Forest models for flood risk	High predictive accuracy; adaptive to dynamic data	Requires large datasets; sensitive to data quality
DL (Xie, 2024)	Image/video analysis for damage assessment	CNNs for post-earthquake structural damage	Automates feature extraction; handles complex data patterns	High computational demands; lacks interpretability
NLP (Godinho, 2024)	Situational awareness from unstructured text	Twitter sentiment analysis during hurricanes	Enables multilingual analysis; real-time information mining	Struggles with sarcasm/ misinformation; language coverage varies
RL (Pu et al., 2025)	Adaptive logistics and routing under changing conditions	Optimized delivery routes during wildfires	Learns optimal actions in real-time scenarios	Computationally intensive; long training time
IoT & AI Integration (A. Khan et al., 2020) (Atzori et al., 2010; Gaur et al., 2021)	Sensor-based disaster detection, infrastructure monitoring	IoT sensors with ML models for early hazard detection	Enables continuous data streams and automation	Vulnerable to cyber threats; requires significant infrastructure (Arfan et al., 2019)

2.3. Relevance of AI-DSS in modern disaster management

As AI technologies advanced, DL and Convolutional Neural Networks (CNNs) were increasingly utilized for image-based disaster assessment, such as damage detection from aerial imagery or satellite data (Sun et al., 2020; Xu et al., 2019). As disasters grow in complexity, the integration of AI into DSS has proven critical. AI-DSS support timely, context-aware decisions by enhancing early warning, logistics coordination, and post-event recovery (Choubin et al., 2019; Cutter et al., 2003; Scawthorn, Blais, et al., 2006; Scawthorn, Flores, et al., 2006; Zerger & Smith, 2003). Recent global events have further underscored the need for AI-DSS. During the COVID-19 pandemic, AI-DSS tools were used for epidemic modeling, financial forecasting, triage planning, and resource distribution (Mammadov et al., 2025; Naudé, 2020; Whitelaw et al., 2020). In wildfire-prone regions like California and Australia, AI-DSS platforms integrated meteorological data, satellite feeds, and vegetation indices to forecast fire spread and guide evacuation efforts (Jain et al., 2020; Liu et al., 2022). Beyond operational advantages, AI-DSS face challenges in interpretability, fairness, and system integration (Abdulrashid, et al., 2023). Addressing these concerns is central to building trust in high-stakes environments. Explainable AI (XAI), human-in-the-loop models, and ethical design are emerging priorities (Barredo Arrieta et al., 2020; Guidotti et al., 2019). Efforts in XAI seek to bridge this gap by making AI models more transparent and interpretable for emergency managers and policymakers (Abdulrashid, Ahmad, et al., 2024; Abdulrashid, Farahani, et al., 2024, 2025; Barredo Arrieta et al., 2020; Guidotti et al., 2019). Without this trust, decision-makers may hesitate to rely on automated outputs (Abdulrashid, Chiang, et al., 2025; Ahmed et al., 2024; Topuz et al., 2025), particularly in high-stakes disaster scenarios.

2.4. Landscape of existing reviews and This Study's Contribution

Prior reviews have addressed individual components of AI in disaster settings, e.g., flood prediction (Mosavi et al., 2018), social media mining (Imran et al., 2015), or wildfire detection (Jain et al., 2020), but few have mapped AI methods to decision-making functions across the entire disaster cycle. Table 3. Limitations of Prior Reviews and This Study's Contribution below summarizes existing limitations.

3. Methodology

This review adopts a systematic, PRISMA-aligned literature review methodology to assess how AI has been integrated into DSS across all phases of disaster management. Rather than focusing on individual AI techniques in isolation, this study emphasizes AI's decision support roles, categorizing the literature by disaster phase, decision function, AI method, data type, and governance implications. Our review was designed to answer five research questions (Section 1) by evaluating existing scholarly contributions,

Table 2
The Evolution of DSS in Disaster Management.

Time Period	Key Characteristics & Technologies	Focus
Pre-2000 s	• Statistical Forecasting • Rule-Based / Expert Systems• Early GIS-enabled DSS	Historical data analysis, Static models
~2000 s-Mid 2010 s	• ML Emergence • Simulation Tools become common • Improved Prediction Models	Better forecasting, Handling structured data
Mid 2010 s-Present	• DL for Image/Sensor Data • AI-IoT Integration • NLP for Text• RL for Optimization • Big Data Analytics	Real-time data integration, Complex patterns
Future	• XAI • Ethical AI Deployment • Hybrid & Autonomous Systems • Integration with Blockchain, 6G, etc.• Advanced NLP (e.g., Disinformation)	Trust, Autonomy, Interdisciplinary Synergy

Table 3
Limitations of Prior Reviews and This Study's Contribution.

Limitation in Prior Reviews	Description	Key Studies Cited
Narrow Focus on Specific Disaster Types	Focus limited to floods, wildfires, or pandemics; lacks generalizability across disaster types.	Mosavi et al. (2018); Jain et al. (2020); Kim et al. (2023)
Emphasis on AI Techniques, Not Decision Context	Reviews describe models (e.g., CNN, SVM) but not how they influence real-world decisions or DSS workflows.	Dargan et al. (2020); Al Marzooqi (2024)
Limited Coverage of DSS Integration	Most studies focus on standalone AI applications without considering their role in integrated DSS platforms.	Albahri et al. (2024); Al Marzooqi (2024)
Insufficient Lifecycle Perspective	Reviews often concentrate on preparedness or response, neglecting recovery and mitigation.	Albahri et al. (2024); Dargan et al. (2020)
Lack of Thematic Classification by Function	Few reviews map AI to specific decision-support functions (e.g., forecasting, triage, logistics, recovery planning).	Imran et al. (2015); Alam et al. (2018)
Neglect of Governance and Explainability Dimensions	Rarely address XAI, human-AI trust, or governance in operational AI-DSS implementations.	Guidotti et al. (2019)
No Unified Conceptual Framework for Researchers	Do not provide structured models or taxonomies that early-career researchers can use to build their own work.	(gap noted across multiple reviews)
Underrepresentation of Hybrid or Man-made Disasters	Reviews typically overlook cyber disasters, civil unrest, or pandemics with mixed causes.	Younis et al. (2023); Miao et al. (2024)

synthesizing their findings, identifying research gaps, and proposing a future-oriented research agenda grounded in operational, theoretical, and methodological needs.

3.1. Data sources and search strategy

A structured, multi-database search strategy was used to ensure a comprehensive and methodologically robust collection of relevant literature. Searches were conducted across leading indexed academic databases, IEEE Xplore, Elsevier (ScienceDirect), SpringerLink, and Web of Science, to capture peer-reviewed and high-quality studies in AI, decision support, and humanitarian logistics. Google Scholar was used only as a supplementary source to identify potentially relevant gray literature or highly cited papers not indexed elsewhere. To minimize potential selection bias, particularly stemming from Google Scholar's broad and non-curated indexing, we implemented several safeguards:

1. Cross-validation: All Google Scholar results were cross-checked against indexed databases. Articles not appearing in at least one indexed database were carefully screened for credibility.
2. Source filtering: Only peer-reviewed journal articles, reputable conference proceedings, and established review papers were retained.
3. Manual screening: Non-scholarly sources, predatory journals, theses, book chapters, and unverifiable publications were excluded.
4. Duplicate removal: Overlapping records across databases were consolidated to avoid overrepresentation of any individual study.
5. Relevance verification: Titles, abstracts, and keywords were reviewed to ensure alignment with AI-driven DSS in disaster or humanitarian logistics contexts.

This multi-step validation process ensured rigor, reduced the influence of Google Scholar's algorithmic ranking, and mitigated database-specific selection bias. In addition to bias mitigation, two independent reviewers screened and assessed all identified records at both the abstract and full-text stages. Reviewer agreement was high, and any discrepancies were resolved through discussion to ensure consistency and reduce subjective bias in the selection process. The final search targeted peer-reviewed journal publications, high-impact conference papers, and authoritative review studies published between 2008 and 2024, reflecting contemporary advancements in AI, digital sensing, and decision support technologies. Search queries combined methodological, domain, and application-specific keywords. Boolean strings included:

- ("Artificial Intelligence" OR "AI" OR "Machine Learning" OR "Deep Learning" OR "NLP" OR "Computer Vision" OR "Reinforcement Learning" OR "Explainable AI")

  AND

- ("Disaster Management" OR "Emergency Response" OR "Crisis Management" OR "Humanitarian Logistics")

  AND

- ("Decision Support System" OR "DSS" OR "AI-DSS" OR "Decision Making").

Additional domain-specific terms such as "smart disaster response," "AI ethics in emergencies," "IoT-enabled monitoring," and "equitable resource allocation" were iteratively added to ensure full coverage. Searches were refined using filters for subject areas

(engineering, computer science, operations management, public safety) and document types (articles, conference papers, review studies). Citation metrics and journal impact factors were also considered to prioritize influential contributions. The overall search and refinement workflow is summarized Fig. 2.

3.2. Inclusion and exclusion criteria

To ensure the rigor, transparency, and relevance of this review, a predefined set of inclusion and exclusion criteria was applied. These criteria were guided by the review's objective to evaluate the integration of AI techniques into DSS for disaster management decision-making across all disaster phases. Eligible studies were selected based on their alignment with the technical scope (AI methods), application domain (disaster management), and contextual focus (decision support). The criteria are summarized in Table 4.

Only studies meeting all inclusion criteria and none of the exclusion criteria were included in the final synthesis. The screening and eligibility process followed a three-step approach:

- Title and Abstract Screening – To exclude studies clearly outside the scope.
- Full-Text Review – To assess methodological and contextual alignment with inclusion criteria.
- Final Selection – Resolved through independent validation by two reviewers, with any disagreements discussed and resolved through consensus or third-party input.

To enhance methodological rigor, we conducted a basic quality appraisal of included studies using a simplified version of the AMSTAR 2 (A Measurement Tool to Assess Systematic Reviews) checklist, adapted for AI and operations management contexts. Each study was evaluated for methodological clarity, data transparency, and validation approach. Studies rated as 'low quality' (e.g., lacking validation or data provenance) were noted but retained to ensure breadth. This appraisal ensured that synthesis prioritized

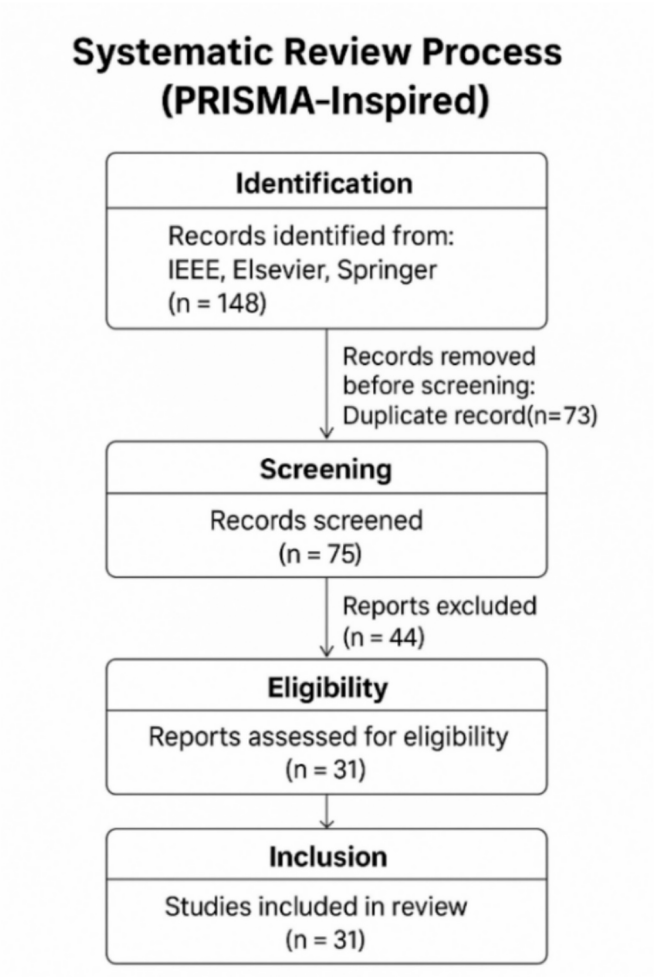

Fig. 2. PRISMA-Inspired Review Flow for AI-DSS in Disaster Management.

Table 4
Summary of Inclusion and Exclusion Criteria.

Criteria Category	Inclusion Criteria	Exclusion Criteria
Scope	Focus on AI techniques applied within disaster management decision-making or DSS frameworks	General ICT tools, visualization platforms, or AI studies unrelated to disasters or DSS
Technology	Studies employing ML, DL, RL, NLP, Computer Vision, IoT-AI, or XAI in a disaster context	Studies discussing AI without implementation or those focusing purely on algorithm development
Functionality	Addresses AI-DSS in disaster preparedness, response, recovery, or mitigation	Lacks a connection to disaster lifecycle stages or practical decision support functions
Document Type	Peer-reviewed journal articles, conference papers, and structured review studies	Editorials, blog posts, theses, technical reports without peer review
Language	English	Non-English publications
Publication Period	Published between 2008 and 2024	Outside of the defined timeframe

studies demonstrating adequate methodological robustness.

Justification of Final Sample Size: Although the initial search retrieved over 2000 documents across multiple databases, only 31 studies met our strict inclusion criteria. The primary reason for this reduction is that a large proportion of AI-in-disaster publications focus on tasks such as hazard prediction, damage detection, or image classification without providing a decision-support function. Because this review specifically targets AI techniques embedded within decision support systems, rather than standalone AI applications, many papers were excluded. Studies lacking explicit DSS components, empirical evaluation, or relevance to humanitarian logistics decision-making were also removed. A confirmatory search conducted during the revision process yielded no additional eligible studies, indicating that the final sample is both methodologically coherent and representative of existing decision-oriented AI-DSS research in disaster and humanitarian logistics.

Coding Procedure: Each included study was coded according to a structured scheme capturing: (1) disaster phase, (2) AI technique, (3) decision function supported, (4) data sources, and (5) validation approach. Coding was performed independently by two reviewers, followed by reconciliation to ensure reliability. This coding matrix formed the basis of the thematic synthesis and framework development.

Quality Assessment: To enhance methodological rigor, each study underwent a quality appraisal using an adapted AMSTAR-2 checklist. The assessment examined methodological clarity, transparency of datasets, appropriateness of the AI technique, evaluation rigor, and relevance to decision support. Studies failing to meet minimum quality thresholds were excluded. This ensured that the synthesis reflects methodologically credible and decision-relevant research.

3.3. Screening and selection process

The screening and selection of studies were conducted using a structured, multi-phase process to ensure relevance, rigor, and transparency. This process followed PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines and is visualized in Fig. 2 below.

- Step 1. Identification: An initial pool of 148 records was retrieved from five major academic databases (IEEE Xplore, Elsevier, Springer, Web of Science, and Google Scholar) using the keyword strategy detailed in Section 3.2. These records included peer-reviewed journal articles and conference papers published between 2008 and 2024.
- Step 2. Screening: After the removal of 73 duplicates and clearly irrelevant records based on title and abstract review, 75 papers remained. Screening focused on topical relevance to AI-DSS and disaster management across the four disaster phases (preparedness, response, recovery, mitigation).
- Step 3. Eligibility Assessment: The remaining 75 full-text articles were examined in detail. A standardized evaluation rubric based on the inclusion and exclusion criteria (see Table 4) was applied. Studies that focused purely on model design without disaster context, lacked decision support application, or did not integrate AI into a broader system were excluded.
- Step 4. Final Inclusion: Following full-text eligibility assessment, 31 studies were included in the final synthesis. These papers were then thematically coded and analyzed according to AI method, disaster phase, decision function, data type, and ethical/governance implications.

Following the inclusion of 31 eligible studies, a structured data extraction process was conducted to enable comparative synthesis and thematic classification. The goal of this step was to capture both the technical attributes of AI methods and their practical roles within DSS implementations across disaster management phases. A data extraction form was developed and pilot-tested on a subset of five papers to ensure clarity and consistency. For each selected study, the following metadata and analytical features were recorded:

- **Bibliographic details**: Authors, year, title, publication type
- **AI method(s)**: ML, DL, RL, NLP, CV, IoT-AI integration, XAI

- **Disaster type**: Natural (e.g., floods, earthquakes), man-made (e.g., pandemics, industrial accidents), or hybrid
- **DSS functionality**: Forecasting, early warning, optimization, coordination, damage assessment, recovery planning
- **Disaster phase**: Preparedness, response, recovery, mitigation
- **Data sources**: Satellite imagery, UAV, sensor networks, social media, health records, environmental monitoring
- **Geographical scope**: Region or country studied (if applicable)
- **Evaluation method**: Case study, simulation, performance metrics, expert validation
- **Ethical/governance dimensions**: Interpretability, transparency, fairness, regulatory context

The extracted data was stored in a structured matrix, allowing for quantitative tabulation and qualitative interpretation. A thematic coding framework was developed to guide the synthesis, categorizing each study across five analytical dimensions: AI Methodology Used; Disaster Phase Targeted; Decision Support Function; Type and Source of Data Utilized; and Ethical and Operational Challenges. Each paper was coded independently by two reviewers, and discrepancies were resolved through iterative discussion. The coding enabled the identification of patterns, clusters, and gaps, highlighting which disaster functions are AI-enhanced, which phases are overrepresented or underexplored, and where novel contributions have emerged. The synthesis was further supported by cross-tabulation matrices (see Section 6) and figures summarizing the distribution of AI techniques across functional and temporal dimensions. These helped identify emerging trends such as the increasing use of deep learning in damage assessment, reinforcement learning in logistics optimization, and XAI frameworks in high-stakes decision environments. This integrative approach ensures that the findings of this review are not only descriptive but also analytically meaningful, offering actionable insights into the current landscape and future potential of AI-DSS in disaster management.

3.4. Textual analysis for thematic validation

To complement the systematic literature review, a textual analysis was conducted on the titles, abstracts, and keywords of the shortlisted studies. Term frequency analysis was used to identify dominant and emerging themes, which were then compared against the SLR-derived categories to validate thematic coverage and reveal topics that are less emphasized in structured coding, such as equity and governance. Fig. 3 presents a word cloud summarizing the results of this analysis. The visualization highlights dominant technical and operational themes (e.g., learning, allocation, decision support) alongside humanitarian and managerial concepts (e.g., logistics, recovery, equity, explainability), while also illustrating the comparatively weaker representation of recovery- and equity-oriented objectives. Collectively, this textual analysis provides complementary evidence that supports the findings of the SLR and helps surface underexplored research directions in AI-driven decision support for disaster and humanitarian logistics.

4. Conceptual framework

To guide our synthesis and anchor this review in operational relevance, we propose a conceptual framework that systematically maps the relationship between AI techniques, decision-making functions, and the phases of disaster management. This framework is designed to go beyond summary, it enables researchers, especially early-career scholars, to identify meaningful intersections, evaluate methodological trends, and propose new research trajectories that are both academically rigorous and practically deployable. Our framework draws from and unifies three well-established domains: (1) Disaster Management, which provides the phase-based structure (preparedness, response, recovery, and mitigation); (2) Operations Management, which informs the decision functions

management
disaster
response
learning
artificial
natural
data
disasters
emergency
intelligence
wildfire
techniques
supply
humanitarian
strategies
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study
models
systems
network
risk
review
preparedness
systematic
various
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recovery
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flood
damage
analysis
predictive
planning
deep
evacuation
prediction
effective
field
paper
monitoring
system
big
model
provides
science
survey
chain
different
hazards
studies
driven
making
gru
post
smart
iot
capacity
events
water
modeling
scenarios
urban
dynamic
proposed
complex
international literature
present
safety
them
social
assessment
analytics
approach
time
level
weather
resource
classification
integration
communication
relief
mesh
benefits
community
problem
how
key
need
impact
neural
access
communities
services
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efforts
aerial
current
results
human
infrastructure
machine
operational
application
fuzzy
leveraging
voice
control
remote
xai
help
fuel
approaches
algorithms
findings
large
method
who
early
computer
phases
involved
messages
contributions
problems
provides
associated
critical
comprehensive
been
cnn
media
rescue
smart
post
iot
capacity
chains
forecasting
principles
new
proposed
complex
dynamic
role
segmentation
organizations
mitigation
first
satellite
imagery
support
technologies
challenges
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such as logistics optimization, infrastructure planning, and resource prioritization; and (3) AI-powered Decision Support Systems, which contribute advanced modeling, data fusion, and real-time automation tools. This three-way synthesis transforms the disaster lifecycle from a static framework into a dynamic system of interdependent decisions, supported by intelligent tools and grounded in real-world implementation needs.

Novelty of the Proposed Framework: While prior reviews in this domain often classify studies by hazard type, algorithm category, or data modality, such taxonomies provide limited insight into how AI techniques support specific decision-making functions within disaster and humanitarian logistics. The framework proposed in this review introduces three conceptual contributions that extend beyond existing literature. First, it incorporates a decision-function layer that explicitly links AI methods to key humanitarian logistics activities, such as demand forecasting, prepositioning, routing, last-mile delivery, and recovery prioritization, an orientation largely absent in earlier reviews. Second, the framework is lifecycle-integrated, mapping decision functions across preparedness, response, mitigation, and recovery, thereby highlighting underexplored intersections such as AI-driven recovery logistics and equity-aware allocation models. Third, it integrates explainability, governance, and ethical considerations as core architectural elements rather than peripheral concerns, reflecting the operational realities and accountability requirements of humanitarian organizations. Finally, by identifying cross-phase AI-DSS pipelines as an emerging research frontier, the framework offers new conceptual insight into how AI components can be connected across stages to enable seamless, adaptive decision-making. Together, these contributions differentiate this framework from previous taxonomies and provide both scholars and practitioners with an actionable, decision-centered lens for navigating the rapidly evolving AI-DSS landscape. The framework additionally distinguishes decision objectives — efficiency, effectiveness, and equity — clarifying how AI techniques operationalize distinct humanitarian priorities across disaster phases.

Fig. 4 illustrates the framework as a layered architecture. At the center are the four canonical phases of disaster management. Surrounding each phase is a ring of key decision functions, including risk forecasting, damage assessment, logistics coordination, and recovery planning, derived from operations research and emergency management literature. Each function is then mapped to relevant AI techniques (e.g., CNNs for image-based damage estimation, reinforcement learning for routing, NLP for real-time awareness). Finally, the outermost layer introduces a contextual overlay, acknowledging the influence of disaster type (natural, man-made, hybrid), data availability, and governance considerations. This layered integration highlights both research concentration (e.g., ML in flood preparedness) and underexplored zones (e.g., equity-focused AI in recovery or RL in pandemics). This framework differs from prior review structures in several key ways. First, it is decision-centered, focusing not only on what AI technologies are used, but why they matter, i.e., which decisions they support, and how they improve disaster outcomes. Second, it is lifecycle-integrated, offering coverage across all four phases rather than narrowly focusing on response or preparedness alone. Third, it incorporates ethics and governance, including transparency, equity, and explainability, which are increasingly critical to real-world adoption of AI in public sector and humanitarian domains. Table 5 summarizes this contribution by comparing our approach to existing review structures in the field. Unlike conventional taxonomies that classify studies by hazard type or algorithm family, our model enables researchers to identify high-impact intersections, such as opportunities to adapt underused AI techniques to underserved disaster phases or to embed transparency into logistics systems. The framework also supports practitioners by clarifying which AI capabilities are most appropriate for different operational decisions, enabling a modular approach to AI-DSS adoption and scaling. This architecture not only structures

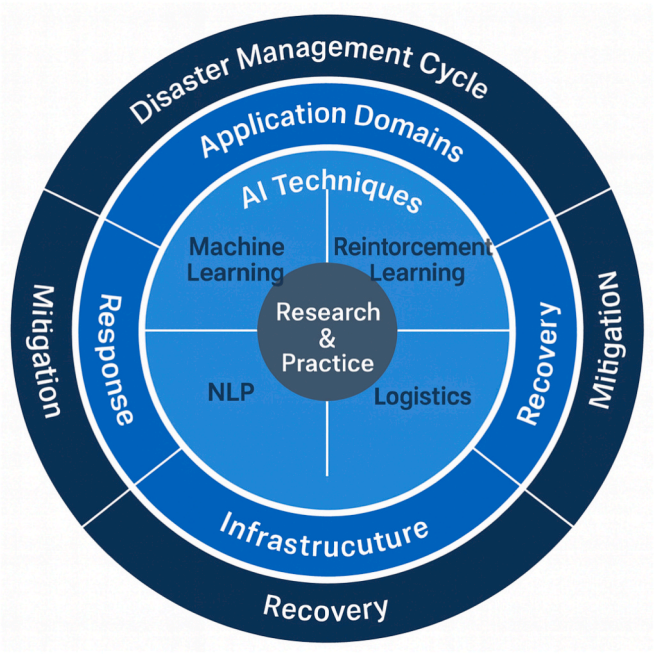

Fig. 4. Conceptual Framework for AI-DSS in Disaster Management.

Table 5
Comparative Review of Prior Literature and This Framework.

Prior Review Focus	Type of Contribution	Decision-Centered	Phase Coverage	Ethics/ Governance	Notable Citation(s)
AI for Flood Prediction	Technical algorithm review	No	Preparedness	No	Pandey (2024); Al-Rawas et al. (2024); Mosavi et al. (2018); Hasanuzzaman et al., (2023); Saravi et al. (2019)
Social Media for Crisis Response	Text mining and NLP applications	Partially	Response	No	Nazer et al. (2017); Nguyen et al. (2016); Zielinski et al. (2013); Imran et al. (2015)
AI in Wildfire Detection	Computer vision and early warning	No	Preparedness	No	Boroujeni et al. (2024); Okoro et al. (2024); et al.; Ramos et al. (2024); Jain et al. (2020)
AI in Humanitarian Logistics	Decision support models for supply chains	Yes	Response	Partial	Christhia et al. (2025); Shakibaei et al. (2024); Florez et al. (2015); Wang et al. (2024); Akter & Wamba (2019)
XAI in Disaster Contexts	Ethical design, XAI reviews	No	All (theory)	Yes	Afroogh et al., (2024); Albahri et al., (2024)
This Review	Decision-function mapping + lifecycle	Yes	All	Yes	This study (2025)

our thematic synthesis in Section 5, but also serves as the foundation for our research agenda in Section 7. It helps transform a fragmented field into a more coherent and actionable body of knowledge, one that combines predictive analytics, operational insight, and responsible innovation to improve disaster resilience globally.

Unlike prior reviews that classify AI applications mainly by hazard type (Mosavi et al., 2018; Akhyar et al., 2024) or algorithm family (Dargan et al., 2020; Albahri et al., 2024), our framework organizes the field by *decision function* (e.g., logistics optimization, forecasting, recovery planning) and *disaster phase*, thereby bridging methodological and operational perspectives. Moreover, this review uniquely embeds ethical and explainability dimensions, aligning with the governance priorities in humanitarian logistics.

5. Thematic synthesis of findings

To deepen the synthesis and address methodological trade-offs across the literature, each subsection concludes with comparative insights that highlight performance patterns, data requirements, limitations, and implications for disaster and humanitarian logistics operations.

5.1. Risk & frequency forecasting

Artificial Intelligence plays a foundational role in forecasting both the likelihood and temporal recurrence of disaster events. These capabilities align closely with preparedness and mitigation decisions (RQ2), and reflect the dominance of supervised learning, remote sensing, and temporal modeling techniques (RQ1). Across the reviewed literature, ML and DL are the most prominent techniques used in predictive systems. To forecast disaster occurrence, severity, and timing, these models ingest longitudinal and real-time data, including satellite imagery, environmental sensor data, and historical hazard records. A summary of the techniques and functions used in disaster management is shown in Table 6.

Risk Forecasting: Risk forecasting centers on estimating future events' probability, location, and intensity. For example, Li et al. (2024) proposed a GRU-CNN hybrid model for real-time flood and landslide prediction using IoT data from smart infrastructure. Similarly, Akhyar et al. (2024) utilized semantic segmentation (U-Net, HRNet) to model spatial wildfire and flood spread using drone and satellite imagery. These models outperform rule-based systems in accuracy and adaptability yet struggle with data sparsity and real-time responsiveness in low-resource regions.

Frequency Forecasting: Disaster frequency forecasting supports long-term planning and resilience-building. AI models trained on historical seismic, meteorological, and healthcare data can predict recurrence intervals of earthquakes, epidemics, and floods. Park and Kang (2024) applied Safety 4.0 analytics to model hazard recurrence zones, while Abid et al. (2021) used AI-GIS fusion to assess landslide and flood frequency across topographies. Miao et al. (2024) advanced this by detecting weak, early signals of unrest and cyber disruptions, highlighting AI's potential in emerging or hybrid threats.

Comparative Insights and Cross-Study Patterns: Across risk and frequency forecasting studies, several methodological patterns emerge. Traditional ML models such as random forests and SVMs tend to perform consistently when training data are structured and moderate in size, offering interpretable feature importance outputs that are useful for operational planning. In contrast, DL architectures, including CNNs, LSTMs, and hybrid GRU-CNN models, often outperform ML in spatial–temporal prediction tasks but require substantially larger datasets and impose greater computational and transparency challenges. Despite performance gains, most models lack uncertainty quantification or decision-oriented output formats, limiting their reliability in humanitarian logistics settings where confidence bounds and scenario trade-offs are crucial. Collectively, the literature suggests that forecasting models are increasingly accurate, but they still struggle with transferability across regions, robustness in data-sparse environments, and interpretability in high-stakes decision contexts.

Table 6
Summary of Techniques and Functions.

AI Technique	Use Case	Data Type	Function
GRU-CNN Hybrid	Flood prediction	IoT sensor streams	Early warning
U-Net, HRNet	Wildfire mapping	Satellite imagery	Spatial spread
Big Data Analytics	Hazard trend modeling	Multiyear archives	Frequency forecasting
Weak Signal Detection	Civil unrest signals	Social & media logs	Emerging threat identification

5.2. Real-Time AI for response & logistics

Disaster response and logistics operations demand rapid, high-stakes decision-making under uncertainty, infrastructure damage, and limited communication. Artificial Intelligence supports this by enabling real-time analysis, dynamic routing, resource prioritization, resilient communication networks, and core tactical and operational DSS functions aligning with the response phase (RQ2). This thematic synthesis draws on AI applications in edge computing, reinforcement learning, NLP, and computer vision (RQ1), highlighting how AI enhances situational awareness, supply chain coordination, and response agility. A summary of some of the key applications is shown in Table 7.

Communication and Coordination in Disaster Zones: Breakdowns in communication infrastructure are a common obstacle during response efforts. Zhou et al. (2021) introduced a DL-based wireless control system that dynamically allocates bandwidth in disaster-affected zones, maintaining connectivity via edge-assisted networks. Similarly, Trotta et al. (2020) employed swarm intelligence with UAV-based mesh networks to restore aerial communications. Lohokare & Dani (2021) implemented opportunistic IoT mesh networks to facilitate last-mile emergency data transmission. Voice-activated AI assistants, as developed by Majumdar et al. (2022), further enhance response coordination by reducing cognitive and manual load. These NLP-based systems support hands-free command execution, real-time query responses, and voice-guided updates for first responders in high-stress settings.

Logistics Optimization and Emergency Routing: Real-time logistics, especially last-mile delivery and supply reallocation, has benefited significantly from AI-driven optimization algorithms. Younis et al. (2023) used machine learning to forecast demand spikes and reroute resources during the COVID-19 pandemic, while Kumar et al. (2025) built a vaccine and PPE delivery model integrating route accessibility, demand density, perishability, and warehouse proximity. These systems adaptively prioritized deliveries under volatile conditions, maximizing impact with limited resources.

AI for Damage Assessment and Navigation: Computer vision systems also assist responders by analyzing aerial imagery to detect impassable roads, collapsed structures, and survivor locations. Gholami et al. (2022) applied CNNs trained on UAV and satellite imagery to classify structural damage and suggest safe navigation paths. Some models integrated thermal imaging and object detection to identify trapped individuals based on heat signatures or motion cues, enhancing triage and rescue prioritization.

Comparative Insights and Operational Implications: In real-time response applications, DL models (e.g., CNN-based damage detection, YOLO architectures for road blockage identification) provide superior performance when used with imagery or sensor-rich environments. However, they are sensitive to noise, require stable connectivity, and are rarely deployable on low-power devices used by humanitarian agencies. Conversely, ML-assisted optimization approaches and hybrid ML–OR models are more robust operationally, though sometimes less precise in rapidly evolving conditions. Reinforcement learning offers strong potential for dynamic routing and resource redistribution but remains under-validated in real deployments due to training instability and data scarcity. A recurring theme is that model sophistication does not necessarily translate to field readiness; simpler or hybrid approaches often provide better alignment with humanitarian operational constraints.

5.3. AI for recovery & infrastructure resilience

The recovery phase of disaster management focuses on restoring disrupted services, rebuilding damaged infrastructure, and supporting long-term socio-economic recovery. AI-powered DSS enhance this process through post-event damage assessment, infrastructure stress forecasting, and resilience-oriented planning. These applications support strategic and operational decisions related to rehabilitation prioritization, zoning, and community reintegration (Summarized in Table 8).

AI for Damage Assessment and Prioritization: Rapid, scalable damage classification is critical for allocating reconstruction resources. Computer vision systems are widely used for this task. Gholami et al. (2022) trained CNNs on xView2 satellite imagery to classify structural damage from earthquakes and floods, enabling tiered response planning. Similarly, Yu et al. (2018) integrated UAV imagery, GIS tools, and social media analysis to produce multi-source impact maps for situational awareness and insurance estimation.

Table 7
Summary of Key Applications.

AI Method	Function	Disaster Need	Example Study
DRL & Edge AI	Bandwidth allocation	Communication resilience	Zhou et al. (2021)
UAV Mesh Swarm	GPS-free comms	Field connectivity	Trotta et al. (2020)
NLP Voice Assistants	Field command automation	Responder coordination	Majumdar et al. (2022)
CNN + UAV Imagery	Damage mapping, routing	Safe access/navigation	Gholami et al. (2022)
ML-based SCM Models	Resource reallocation	Last-mile delivery	Younis et al. (2023)

Table 8
Summary of Key Applications.

AI Method	Function	Recovery Use Case	Example Study
CNN on Satellite/UAV	Damage classification	Infrastructure triage	Gholami et al. (2022)
GRU-CNN + IoT	Structural stress modeling	Collapse risk prediction	Li et al. (2024)
AI-GIS Tools	Zoning & reconstruction plans	Resilience design	Abid et al. (2021)
Policy Simulators	Sea-level risk mapping	Long-term planning	O'Brien et al. (2006)

These systems significantly reduce manual inspection time, prioritize infrastructure recovery, and support claims processing, especially in remote or severely damaged zones.

Forecasting Infrastructure Failure and Recovery Bottlenecks: Beyond static assessments, AI is increasingly used for simulation-based resilience forecasting. Li et al. (2024) used GRU-CNN hybrids to simulate infrastructure stress under seismic and flood conditions. Their smart city system predicted high-risk collapse zones using IoT sensor data and material metadata, supporting decisions on retrofitting, structural audits, and emergency sheltering. Simulation-enhanced DSS also help governments model supply chain recovery, utility restoration, and temporary housing deployment timelines, optimizing phased resource allocation.

Policy Planning and Risk-Informed Urban Design: Long-term recovery requires AI tools that support resilience-by-design policies. Abid et al. (2021) demonstrated the use of AI-GIS platforms to develop zoning maps that incorporate environmental, seismic, and demographic risk layers. O'Brien et al. (2006) integrated AI with climate projection data and land-use modeling to inform adaptive policies, especially for coastal hazard mitigation and reconstruction.

Equity-Focused Recovery Support: Emerging AI-DSS systems also include social vulnerability indices, such as housing density, healthcare access, and poverty rates. These indicators help prioritize rebuilding in underserved areas, address post-disaster inequities, and guide inclusive aid distribution. However, few models currently combine infrastructure risk with demographic equity factors in an integrated decision pipeline.

Comparative Insights and Observed Gaps: Compared to preparedness and response, recovery-focused AI-DSS research is sparse and fragmented. Most existing tools rely on ML classifiers or static optimization models, which are insufficient for modeling long-term infrastructure interdependencies or community-level recovery trajectories. Simulation and agent-based models offer richer representational power but are rarely integrated with modern AI techniques. Cross-study comparison indicates that recovery-phase AI research is limited by data scarcity, the absence of longitudinal ground truth, and the complex interplay between social, economic, and physical systems. This gap underscores a strategic opportunity for developing hybrid, interpretable models capable of supporting equitable and transparent recovery decisions.

5.4. Human-centered AI: NLP, equity, and community use

Human-centered AI-DSS applications focus on bridging the gap between raw data and actionable, socially responsive decision-making. These systems leverage unstructured text, voice, and crowd-sourced data to enhance situational awareness, streamline communication, and address equity in disaster management. Unlike purely environmental or structural modeling, this domain emphasizes real-time public insights, human-AI interfaces, and inclusive system design (summarized in Table 9).

NLP for Situational Awareness and Misinformation Tracking: NLP enables disaster managers to analyze social media, news, and dispatch logs in real time. Imran et al. (2015); Alam et al. (2018) demonstrated how AI models extract crisis-related information, such as shelter needs, hazard zones, and public sentiment, from platforms like Twitter and emergency call transcripts. These insights have been used for mapping emergent hotspots, prioritizing rescue or supply routes, and monitoring evolving misinformation trends. However, models must navigate challenges in language diversity, sarcasm detection, and misinformation spread, especially in politically sensitive or linguistically diverse regions.

Voice Interfaces and Edge NLP for Field Operations: To support hands-free, intuitive interaction in chaotic environments, Majumdar et al. (2022) developed an AI-powered voice assistant using speech-to-text NLP models. This system allows responders to issue verbal commands, update status, or request aid without pausing critical fieldwork, improving safety, reducing manual error, and enabling low-literacy or multilingual use. When paired with edge computing, such systems enable offline or low-bandwidth operations, which are especially valuable in remote or conflict-affected regions.

Crowdsourcing and Participatory AI: Participatory platforms that incorporate citizen-generated data, via SMS, voice, or mobile apps, can inform local-level decisions that top-down systems often overlook. For example:

- SMS-based systems used during earthquakes or floods to geolocate needs
- Community surveys for damage reporting and aid tracking
- Apps enabling displaced persons to register or request support

Despite the potential, few AI-DSS models formally integrate this "bottom-up" data stream into predictive models or dashboards. The absence of data validation protocols, privacy protections, and standardized formats limits their current use in operational pipelines.

Table 9
Summary of Key Applications.

AI Technique	Function	Application Domain	Example Study
NLP	Real-time awareness, sentiment mining	Crisis mapping, misinformation detection	Imran et al. (2015); Alam et al. (2018)
Voice-AI	Command & control in-field	Hands-free responder interfaces	Majumdar et al. (2022)
Crowd AI	Participatory damage reporting	Equity-aware needs mapping	Yu et al. (2018); WHO (2023)
XAI + Equity-AI	Trust, transparency, fairness	Human-AI collaboration frameworks	Abdulrashid et al. (2024)

Table 10
Summary Table of Cross-Phase Use Cases.

Function	Linked Phases	AI Methods	Example Studies
Risk-informed logistics	Preparedness → Response	GRU-CNN + RL	Li et al. (2024)
Damage feedback loop	Response → Recovery → Mitigation	CNN + Big Data Fusion	Yu et al. (2018); Gholami et al. (2022)
Infrastructure stress prediction	Preparedness → Recovery	IoT + Transfer Learning	Jain et al. (2020); Xu et al. (2019)
Policy modeling	Mitigation → Preparedness	Simulation + Scenario Forecasting	Chang et al. (2024); O'Brien et al. (2006)

Ethical and Inclusive Design Considerations: Human-centered AI in disaster contexts raises urgent issues around:

- Bias amplification (e.g., overrepresenting digitally connected groups)
- Accessibility (e.g., for the elderly, disabled, or linguistically isolated)
- Transparency and consent in AI-enabled decision-making

Efforts in XAI and ethical AI design are emerging to improve user trust and accountability. Yet, few models are co-designed with end-users, particularly in the Global South or in humanitarian contexts.

Comparative Insights and Human Factors: NLP-based systems provide strong capabilities for extracting situational signals from crowdsourced text data, yet they face challenges related to linguistic bias, message noise, and domain drift across regions. Compared to structured ML models, NLP and human-centered AI systems require significantly stronger attention to usability, trust, and ethical alignment. The literature shows a consistent gap in end-user validation: few studies evaluate how emergency managers interpret AI outputs or how interface design affects decision quality. Equity-aware modeling remains underdeveloped despite its critical importance in humanitarian contexts. This highlights the need for interdisciplinary approaches that integrate social science, HCI, and ethics into AI-DSS design.

5.5. Cross-phase AI pipelines

While many AI applications in disaster management are confined to specific phases, e.g., risk forecasting during preparedness or logistics during response, a growing number of efforts aim to build cross-phase AI-DSS pipelines that support continuity and adaptability across the entire disaster lifecycle (Summarized in Table 10). These pipelines integrate predictive, monitoring, optimization, and recovery tools into a modular, dynamic decision support environment, enabling:

- Seamless transition between early warning and operational response
- Feedback loops from post-disaster data to improve future mitigation strategies
- Resource optimization across both emergency and long-term planning contexts
- Integrated AI-DSS Architectures

Li et al. (2024) proposed a smart city architecture where sensor-based risk forecasts (preparedness) inform evacuation planning (response) and structural health monitoring (recovery). This system continuously ingests real-time data from urban infrastructure, offering city officials a unified dashboard for disaster risk and resilience management. Similarly, Yu et al. (2018) introduced a big-data-driven DSS capable of drawing on satellite imagery, IoT feeds, and crowd reports. Their system supports risk analysis, logistics coordination, and post-disaster assessment, demonstrating a practical implementation of multi-phase data fusion.

Feedback-Driven Resilience Loops: Cross-phase systems increasingly emphasize feedback-driven learning, wherein data gathered during one phase enhances decisions in future events. For instance:

- UAV-based damage assessment (recovery) can update hazard models for future preparedness.
- Mobility patterns tracked during evacuations can train RL models for better routing under future crises.
- Recovery prioritization and reconstruction data feed long-term mitigation simulations (e.g., urban planning or infrastructure hardening).

Table 11
Mapping Objective Functions, AI Techniques, and Disaster Phases.

Objective Function	Operational Focus / Metrics	AI Techniques Commonly Used	Disaster Phases Applied	Representative Decision Tasks
Efficiency	Cost minimization, distance reduction, resource utilization	ML, RL, Optimization-based AI	Preparedness, Response	Facility location, routing, inventory prepositioning
Effectiveness	Delivery time, coverage, service level, deprivation reduction	ML, DL, NLP	Response, Recovery	Demand forecasting, last-mile delivery, damage assessment
Equity	Fairness, vulnerability prioritization, access parity	NLP, XAI, MCDM, Hybrid AI-OR models	Response, Recovery	Equitable allocation, needs assessment, recovery planning
Robustness	Resilience under uncertainty, adaptability to disruptions	RL, Bayesian ML, Simulation-AI	Preparedness, Response, Recovery	Dynamic rerouting, adaptive resource allocation
Transparency	Interpretability, accountability, trust	XAI (SHAP, LIME), Rule-based AI	All phases (decision oversight)	Model explanation, justification of logistics decisions

This feedback mechanism represents a key step toward adaptive governance, where systems improve over time through cumulative learning.

Modular vs. Monolithic Pipelines – A critical architectural choice is whether to develop:

- Modular systems, which integrate independent AI tools through common APIs or data standards (more flexible, but harder to synchronize); or
- Monolithic platforms, where a centralized engine handles prediction, optimization, and communication (easier to scale, but harder to tailor).

Current best practices suggest hybrid architectures, where modular AI models (e.g., NLP, CV, RL) are embedded within standardized DSS shells using GIS, web interfaces, or cloud dashboards.

Mapping objective functions to AI techniques reveals a strong efficiency bias in existing AI-DSS research, with comparatively limited attention to equity-driven objectives. NLP and XAI methods show particular promise for addressing fairness and inclusion, especially in the response and recovery phases, yet remain underutilized. Table 11 synthesizes how AI-DSS studies operationalize key objective functions —efficiency, effectiveness, equity, robustness, and transparency—by mapping them to AI techniques and disaster phases, thereby highlighting both dominant research patterns and underexplored humanitarian priorities.

Comparative Insights and Lifecycle Integration: Only a small subset of studies attempt to integrate AI techniques across multiple disaster phases, and these efforts remain largely conceptual. Modular ML pipelines provide flexibility but struggle with real-time disruptions, while end-to-end DL approaches offer efficiency at the cost of interpretability and data dependency. Cross-study patterns reveal that current AI-DSS are siloed in terms of both phase focus and decision function, inhibiting their ability to support the multi-stage, interdependent nature of humanitarian logistics. This underscores a major research gap: the need for lifecycle-integrated AI-DSS capable of combining forecasting, routing, allocation, and recovery into a coherent, adaptive architecture.

6. Data and modeling considerations

6.1. Data sources and accessibility (Aligned with RQ3)

Data remains the foundation for all AI-powered DSS applications in disaster management. However, the availability, structure, and accessibility of data vary significantly across disaster types and application domains, creating substantial disparities in the maturity and applicability of AI models. Studies of natural disasters, particularly floods, earthquakes, and wildfires, benefit from decades of archived remote sensing, seismic, and meteorological data collected by global organizations and satellites. Structured datasets from agencies such as NASA, EM-DAT, and Copernicus facilitate supervised learning approaches and consistent benchmarking. This abundance is reflected in the concentration of AI-DSS research in domains like flood forecasting, wildfire detection, and structural damage classification (Jain et al., 2020; Mosavi et al., 2018). In contrast, man-made and hybrid disasters, such as pandemics, cyberattacks, and civil unrest, suffer from limited structured data. These domains often rely on unstructured or semi-structured sources such as social media streams, electronic health records, and government reports, data types that are either hard to access, lack labeling, or require significant preprocessing. Studies like Imran et al. (2015) and Miao et al. (2024) have demonstrated early success in using these sources for real-time analysis, but widespread adoption remains limited due to noise, privacy concerns, and lack of standardization. Access to proprietary and non-open data, such as telecom metadata, emergency call transcripts, or private drone footage, further complicates the picture. While potentially rich in information, these datasets are rarely shared due to legal, ethical, or commercial constraints, reducing the reproducibility of AI-DSS models and limiting their scalability in real-world systems. The chart in Fig. 5 provides a visual comparison of AI-DSS research coverage across disaster types and functional domains, highlighting how data accessibility directly correlates with research intensity.

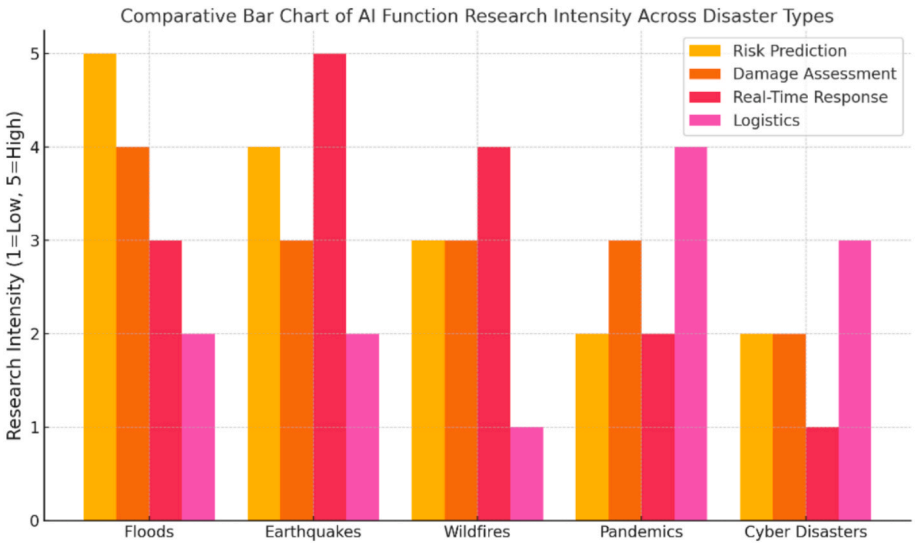

Fig. 5. Comparative Bar Chart of AI Function Research Intensity Across Disaster Types.

6.2. *Emerging modeling techniques (Aligned with RQ1 & RQ3)*

The modeling techniques used in AI-DSS for disaster management are evolving rapidly, from traditional machine learning to increasingly complex and specialized frameworks. While early work relied heavily on supervised learning (e.g., Random Forest, SVM, Decision Trees), more recent studies have shifted toward advanced techniques such as deep learning, reinforcement learning, and hybrid neural architectures capable of handling multimodal and real-time data.

Supervised and Deep Learning Models: Most current implementations still use CNNs and recurrent neural networks (RNNs) for tasks such as image classification (e.g., building damage detection from satellite imagery) or time-series forecasting (e.g., flood or seismic activity). These models perform well in settings with rich, labeled datasets, especially for natural disasters where remote sensing and sensor data are abundant (Gholami et al., 2022; Li et al., 2024). However, these models often act as black boxes, making them difficult to interpret, particularly in high-stakes settings where decision-makers need to justify actions. As a result, there's growing interest in integrating explainability layers (e.g., SHAP, LIME, FIRM) or using more interpretable models that can offer trade-offs between accuracy and transparency.

Reinforcement Learning and Real-Time Optimization: For decision-making in dynamic environments (e.g., evacuation routing, resource allocation), RL is gaining traction. Studies such as Zhou et al. (2021) and Younis et al. (2023) demonstrate how RL can dynamically optimize logistics and communication pathways during rapidly evolving crises like pandemics or wildfires. However, RL models typically require high volumes of training episodes and struggle in sparse-reward environments, limiting their use in rare-event disaster contexts without robust simulation environments.

Hybrid and Ensemble Models: To overcome the limitations of individual models, ensemble methods and hybrid architectures are increasingly being adopted. For instance, GRU-CNN hybrids can process both spatial and temporal data streams, enabling enhanced performance in forecasting cascading disaster events. Miao et al. (2024) introduced weak signal modeling for early detection of cyber-social unrest, highlighting how hybrid models can detect low-frequency, high-impact patterns often missed by conventional approaches.

6.3. *XAI and governance integration (Aligned with RQ1, RQ4, and RQ5)*

As Artificial Intelligence becomes embedded in high-stakes disaster decision-making, the need for transparent, interpretable, and ethically governed AI models, collectively referred to as XAI, has become critical. While AI-DSS tools offer powerful predictions and optimization, their adoption in practice hinges on the trust and understanding of emergency managers, policymakers, and affected communities.

Why Explainability Matters in Disaster Contexts: In disaster response scenarios, where decisions affect lives and infrastructure in real time, opaque “black-box” predictions from deep learning models are often insufficient. Stakeholders must be able to trace the rationale behind AI recommendations, whether for evacuation zones, supply allocation, or damage estimates. Without transparency:

- Emergency responders may hesitate to act on AI output.
- Legal and policy frameworks may reject liability tied to automated decisions.
- Affected populations may resist interventions seen as unjust or misaligned.

Current State of XAI in AI-DSS: Despite growing awareness, few reviewed studies integrate explainability systematically. SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) have been used for post-hoc model auditing in risk prediction and severity estimation (Ghaffarian et al., 2023). FIRM (Feature Importance Ranking Methodology) has emerged as a more robust method for quantifying which features (e.g., wind speed, infrastructure density) most influenced AI outputs, aiding interpretation for planners and domain experts. Visual and verbal tools like saliency maps or natural language model summaries are promising but remain underutilized in field-deployed systems. Still, most explainability tools are used in academic post-processing, not in real-time operations. This gap presents a barrier to human-in-the-loop DSS design.

Governance, Ethics, and Accountability: Beyond technical interpretability, AI-DSS must be governed by frameworks that ensure fairness, accountability, and legal compliance. Key concerns include:

- Bias propagation from historical or skewed datasets (e.g., resource allocations reflecting systemic inequalities).
- Privacy concerns when integrating health records, mobile location data, or social media.
- Autonomy vs. automation, especially in contexts like triage or resource denial during pandemics.

As Barredo Arrieta et al. (2020) and Guidotti et al. (2019) emphasize, XAI must be coupled with institutional design, including audit mechanisms, model documentation, and ethical review boards. Frameworks like the EU AI Act and OECD Principles on AI offer early models for responsible deployment.

Modeling trade-offs and real-world applicability

Beyond methodological considerations, the operational feasibility of AI-DSS remains a significant challenge. Many high-performing models rely on large, high-quality datasets that are seldom available during disasters, particularly in low-income regions. Deep learning architectures require computational resources and connectivity that may not exist in field settings, while simpler models, although more robust, may fail to capture nonlinear patterns or evolving hazard dynamics. Multimodal pipelines integrating imagery, text, and sensor data offer richer situational awareness but introduce greater complexity and governance concerns related to data access, privacy, and interoperability. These constraints underscore the importance of designing AI-DSS that are not only technically sophisticated but also deployable under real humanitarian logistics conditions: low bandwidth, intermittent power, uncertain data quality, and multi-agency coordination demands. Addressing these real-world constraints remains a critical direction for future research.

7. Research gaps and future directions

The systematic review highlights substantial opportunities for advancing AI-driven decision support in disaster management and humanitarian logistics. However, the current research landscape remains fragmented across methodological, operational, and ethical dimensions, limiting both theoretical development and real-world implementation. To provide a clearer structure and a more strategic research agenda, we organize these gaps into three thematic categories: methodological, operational and implementation, and ethical and societal gaps. This organization helps illuminate how each domain shapes the future trajectory of AI-DSS research and informs the design of resilient humanitarian supply chains.

7.1. Methodological gaps

Methodological limitations constitute one of the most significant barriers to progress in AI-DSS for disaster and humanitarian logistics. Although numerous studies propose models for forecasting hazards, routing supplies, or analyzing damage, very few offer integrated approaches that span the full disaster lifecycle. Most AI applications are tightly scoped to preparedness or response, leaving recovery and mitigation, phases that involve long-term planning, reconstruction, and resilience building, largely unaddressed. This lack of lifecycle integration restricts the utility of current systems for holistic decision-making. Another methodological challenge relates to the limited use of models capable of handling uncertainty. Disaster environments are inherently stochastic, yet many AI-DSS rely on static, supervised learning methods that assume stable data distributions. Advanced approaches such as Bayesian learning, uncertainty-aware prediction, reinforcement learning, or hybrid simulation–optimization frameworks are rarely used, even though they are well-suited to dynamic and resource-constrained environments. At the same time, the growing availability of multimodal data, including satellite imagery, social media text, and IoT sensor streams, has not been matched by methodological innovation. Few studies integrate multiple data types within a single decision pipeline. Likewise, the generalizability of existing models remains limited, with most systems validated only in small case studies or local settings. This narrow methodological focus highlights the need for more robust, scalable, and multimodal approaches that can adapt to diverse humanitarian contexts.

7.2. Operational and implementation gaps

In addition to methodological constraints, significant operational gaps hinder the translation of AI-DSS into real-world disaster and humanitarian settings. Current systems often function as isolated analytical tools rather than components of broader operational workflows. There is limited attention to interoperability between forecasting models, routing systems, and allocation tools, resulting in fragmented decision landscapes that mirror institutional silos rather than integrated operations. A related challenge is the limited incorporation of human expertise into AI-DSS design. Most systems are not evaluated with disaster managers, humanitarian workers, or local community stakeholders. Without integration of human feedback and participatory design, even high-performing models may

fail to gain operational trust or align with field realities. Moreover, practical deployment considerations, such as the need for low-connectivity operation, computational robustness during infrastructure disruptions, or compatibility with existing humanitarian logistics platforms, are rarely addressed. Cross-agency coordination also remains underexplored. Disaster response typically involves collaboration among governments, NGOs, international relief agencies, and private sector logistics partners. However, the literature provides few examples of AI-DSS designed to facilitate data sharing, coordination, or collective decision-making across institutional boundaries. Consequently, operational improvements achieved by AI-DSS at the local model level often fail to translate into systemic gains.

7.3. Ethical and societal gaps

Ethical considerations are increasingly prominent in AI research, yet they remain peripheral in the context of disaster and humanitarian logistics. Many existing systems rely on datasets that underrepresent vulnerable communities, leading to biased predictions or inequitable resource allocation. Despite the growing recognition that humanitarian logistics decisions often have ethical implications, few studies incorporate equity-aware or fairness-oriented modeling. Transparency and explainability also remain major challenges, particularly for deep learning systems that offer predictive power but limited interpretability. In high-stakes disaster environments, where accountability and trust are critical, opaque models present serious obstacles to adoption. Another underdeveloped area is community participation: affected populations are rarely involved in designing or evaluating AI-DSS, despite having unique contextual knowledge that could improve system relevance and legitimacy. Data governance further complicates the ethical landscape. Concerns about privacy, secure data sharing, and responsible use of sensitive information are seldom acknowledged, even though humanitarian data often involve personally identifiable information or politically sensitive geospatial records. Addressing these concerns is essential for building trustworthy AI-DSS that align with humanitarian principles.

7.4. Linking research gaps to future opportunities

Collectively, these methodological, operational, and ethical gaps point to a rich landscape of future research opportunities. To translate these gaps into actionable pathways, Table 12 outlines representative research questions, suitable methodological approaches, and potential data sources that correspond to each thematic area. By framing future work in terms of concrete research tasks, such as integrating multimodal data for real-time logistics optimization, developing interpretable reinforcement learning models, or incorporating equity metrics into recovery planning, Table 12 provides a practical roadmap for advancing both theory and practice.

7.5. Strategic priorities for the field

Although numerous opportunities exist, several cross-cutting priorities stand out as especially urgent. First, the development of integrated AI-DSS architectures capable of bridging early warning, response, and recovery is essential for building resilient humanitarian supply chains. Second, human-centered and explainable AI must become core design principles to ensure that systems are transparent, trustworthy, and usable by practitioners. Third, increased focus on equity-aware modeling is necessary to ensure that AI-driven decisions do not inadvertently reinforce existing vulnerabilities or inequalities. Finally, sustained progress will depend on collaboration across disciplines and institutions, enabling the co-creation of AI tools that reflect both technological advances and the realities of humanitarian operations.

8. Conclusion

This review examined the role of AI-DSS in enhancing disaster operations across the preparedness, response, recovery, and mitigation phases. Drawing on more than thirty peer-reviewed studies, the findings demonstrate that AI-DSS have become critical enablers of faster, more informed, and more coordinated decisions under conditions of high uncertainty and resource scarcity. AI methods, including machine learning, deep learning, natural language processing, reinforcement learning, and AI–IoT integration, support diverse decision tasks such as risk forecasting, logistics optimization, damage mapping, and recovery planning. These advances extend the capabilities of traditional DSS beyond static dashboards, providing opportunities for real-time situational awareness and adaptive response. Despite these advances, deployment in practice remains limited. Most AI-DSS applications remain experimental, with minimal integration into institutional workflows or cross-agency platforms. Persistent challenges include fragmented system architectures, lack of transparency in high-stakes contexts, limited data quality in under-resourced regions, and narrow applicability to non-natural disasters such as cyberattacks and cascading infrastructure failures. Addressing these challenges will be essential to realize the promise of AI-DSS in disaster and humanitarian logistics.

8.1. Theoretical contributions

This review makes several contributions to the growing intersection of AI, decision support, and humanitarian logistics. First, it introduces a decision-oriented conceptual framework that systematically maps AI techniques to logistics decision functions and disaster phases. This framework goes beyond algorithmic classification, highlighting how AI supports structured decision processes in humanitarian supply chains. Second, the synthesis identifies research gaps in recovery, mitigation, and equity-focused modeling, thereby extending the theoretical boundaries of humanitarian logistics, which has traditionally emphasized preparedness and

Table 12
Sample Research Questions and Approaches for Future AI-DSS Studies.

Research Question	Suggested Methodologies	Potential Data Sources
How can AI-DSS balance predictive accuracy with transparency in high-stakes response contexts?	SHAP + scenario-based simulation	xView2, EM-DAT, FEMA case reports
What hybrid models can integrate social media and IoT sensor data for situational awareness in cyber-disasters?	Multimodal ML + NLP + graph learning	CrisisNLP, Twitter APIs, sensor networks
How can recovery-focused AI-DSS incorporate equity metrics into rebuilding prioritization?	Multi-criteria optimization, participatory modeling	OpenStreetMap, UNDP recovery datasets, census data
What design features increase trust and usability of XAI interfaces among disaster managers?	Mixed methods (interviews + prototyping)	End-user testing, survey data, usability logs
Can reinforcement learning optimize multi-phase logistics across dynamic disaster scenarios?	Deep RL + agent-based modeling	Historical supply data, humanitarian logistics archives

response. Third, by integrating insights from operations management, disaster studies, and AI research, the review provides an interdisciplinary lens that enriches the theoretical foundations of both AI-DSS and humanitarian logistics literature.

8.2. Practical implications

For practitioners, especially those involved in humanitarian supply chains, this review highlights pathways for adopting AI-DSS to improve operational effectiveness. Opportunities include:

- Strategic: AI-enabled facility location and inventory prepositioning models to strengthen preparedness.
- Tactical: Reinforcement learning and optimization tools to improve transportation routing, last-mile delivery, and equity-aware allocation during crisis response.
- Operational: IoT-enabled monitoring and real-time analytics to support situational awareness, recovery prioritization, and adaptive logistics planning.

These insights can guide humanitarian organizations, NGOs, and governments in investing in AI solutions that augment, rather than replace, human judgment. Equally important are governance and policy mechanisms, such as explainable AI frameworks, cross-sector data sharing protocols, and participatory design, that ensure adoption in real-world relief settings.

8.3. Limitations

While comprehensive, this review has limitations that open opportunities for future research. The analysis is restricted to studies published between 2008 and 2024, potentially excluding recent work emerging in fast-moving AI domains. The reliance on peer-reviewed literature may underrepresent practical insights from field deployments, NGO reports, or proprietary systems. Furthermore, while the review emphasizes the integration of AI into humanitarian logistics decision-making, many applications remain proof-of-concept, with limited empirical validation in real-world crises. Finally, the scope of reviewed AI methods is broad but not exhaustive, and emerging technologies such as generative AI, blockchain integration, and edge-AI systems warrant deeper exploration.

CRediT authorship contribution statement

Ismail Abdulrashid: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Mohamed Khalafalla:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Nikhil Sarma Kadiyala:** Writing – original draft, Visualization, Software, Resources, Investigation, Formal analysis. **Wen-Chyuan Chiang:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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